Hobacare interface docking V2. 2.3

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1. Change Notes

Revision Date	Change Type	Revisions	Remark
2023-12-18	Added	Added whitelist management	
2024-03-27	Revision	Adjust the document format and add more detailed description content	
	Added	Added https://qinglanst.com/prod-api/thirdparty/v2/subOwner 1. Subscribe to events of all devices in the enterprise without specifying UID 2. Added signal anomaly and angle anomaly event subscription It is recommended to use this interface instead of the old interface https://qinglanst.com/prod-api/thirdparty/v2/subAffair	
	Revise	https://qinglanst.com/prod-api/thirdparty/v2/prop Supports breathing sleep mode settings, and adds restrictions on device models and modes	
		https://qinglanst.com/prod-api/thirdparty/v2/bindDevice When the binding is successful, the return value adds "online status" and "device model" information	
		https://qinglanst.com/prod-api/thirdparty/v2/prop Support setting breathing sleep mode, add offline and scene incompatible status codes	
		https://qinglanst.com/prod-api/thirdparty/v2/prop https://qinglanst.com/prod-api/thirdparty/v2/border https://qinglanst.com/prod-api/thirdparty/v2/coords https://qinglanst.com/prod-api/thirdparty/v2/threshold Add offline status code	
		https://qinglanst.com/prod-api/thirdparty/v2/getDeviceInfo When getting the device list, the return value adds device type information	
		The TCP connection message length is limited to 32KB or less (if the message length does not exceed 32K bytes, it will not be affected by this requirement)	
	No longer maintained	https://qinglanst.com/prod-api/thirdparty/v2/subAffair	The existing interface is no longer maintained, but can still be used
2024-05-01	Added	setBreathHeartParam interface adds suddenDuration and intensiveCare parameters (supported by the firmware in April)	April firmware support
	Added	Added app_compile_time and radar_compile_time properties to getDeviceProp interface	
	Added	TCP subscription second-level trajectory and breathing heart rate (limited use) added code	
2024-05-22	Added	Added large screen ranking interface getRiskRanking (get ranking version)	
	Revise	Modify the online and offline push logic: To reduce the alarm push caused by network fluctuations, no online and offline notification will be generated if the device goes online within 1 minute after being offline.	
	Revise	getSleepReport (Get Sleep Report) interface current limit adjustment	
2024-06-03	Added	Added sensitivity parameter to setBreathHeartParam interface (supported by firmware in April)	April firmware support
2024-06-20	Added	MQTT pushes real-time trajectory, breathing and heart rate	June firmware support
	delete	Remove the old TCP interface (the old interface can still be used), and recommend using MQTT push	
	Revise	setDeviceProp (Set Basic Properties) adds new parameters for bed leaving, bed retention, and long-term absence detection subscribeAffair (Subscribe to Events) Added bed leaving, staying, and long-term unattended detection	
	Revise	Sleep report adds daytime trajectory information	

2024-07-15	Added	Added jump to whiteboard applet monitoring page	
	Added	Added getWxToken interface to obtain mini program token	
2024-08	Additional Notes	Added example description in Section 2.4.1; added description of automatic measurement boundary setting in Section 4.2.3; added description of “warning area” definition in Section 4.2.4; added description of sleep state definition in Section 5.2	
	Added	Section 2.5 Added web page configuration and query callback interface functions 4.1.4 /4.1.5 Added event subscription and callback for "enter and exit alarm area"	
2024-10	Revision	The directory structure has been changed, and the mini-program network configuration and device call jump functions have been added; the Web call device jump function has been added; and the MQTT connection self-service configuration function has been added.	
2024-12	Revision	Delete Section 4.3 and merge it into Section 4.2: Provide H5 page embedding, suitable for mini-programs and APP development.	
2025-01	Revision	2.3.2/5.2.4 New event 9 Add more alarm type descriptions 3.3.1 Added definitions for postures related to sitting on the floor and sitting up on the bed 5.3.1 Added a new parameter for sudden death posture after leaving the bed for a long time 5.3.2 Added a new parameter for sudden death posture 5.5.1 Added a new optional parameter version The latest sleep report added eye movement status; added Chinese and English examples	

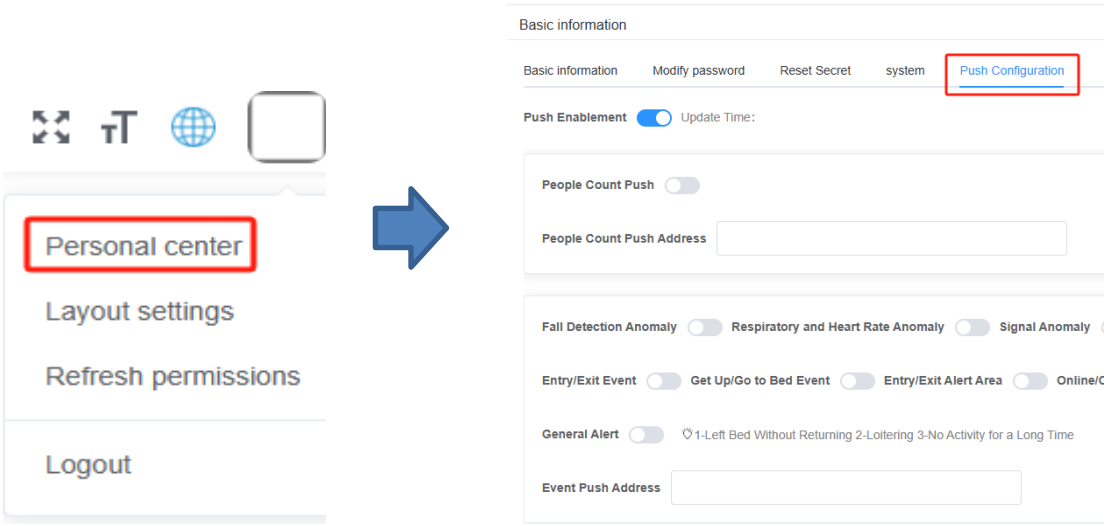
2. Quick docking event notification

2.1. Get a Hobacare platform account

Before docking, you need to ensure that the company has registered on the Hobacare platform and obtained an account and password. If not, please ask the company administrator to contact Hobacare technical sales staff to create the company and provide account information.

2.2. Register HTTP callback interface

Hobacare platform using the Enterprise account and password in 2.1 and register the event callback URL as shown in the figure below: **Personal center ☒ Push Configuration**



The push enable button status indicates whether push is currently in progress. If it is gray, push is stopped.
The other buttons indicate whether to push this event/alarm. The push address indicates the address to which the event/alarm is pushed. The Hobacare system currently supports setting two URLs:

- Number of people push address: URL for receiving notifications of changes in the number of people
- Event push address: URL for receiving device alarms or events

- Please be sure to test whether the URL is accessible on an external network (or using a mobile phone 4G). If you cannot receive events, it is likely caused by a firewall or LAN.
- If the server where your URL is located is down for more than 2 hours, push will be stopped. You can use the above process to check whether the current push is enabled on the Hobacare platform and re-enable it. It is recommended to use multi-threaded processing of the interface.
- After the settings are completed, it is recommended to use the number of people or online and offline events to test whether the event is pushed correctly. Note that the online and offline events will only be sensed and triggered if the device is powered off for more than 3 minutes.

2.3. Install equipment and debug events

Refer to the following Hobacare equipment manual and installation instruction video to install and debug the equipment.
Please ask the sales to send you related documents.

After successful debugging, the events generated by the device will be pushed through the URL registered in 2.2.

The following are the events currently supported by the system:

Event value	Event Type	Event Level
1	Changes in the number of people	Common events
2	Fall incidents	Alerts
3	Respiration and heart rate related events , the meaning of the event is determined by the params parameter: 11: Hyperpnea 12: Hypopnea 13: Apnea 14: Heart rate is too high 15: Heart rate is too low 16: Vital signs are weak	Weak vital signs are warnings; others are common events
4	Door entry and exit events	Common events
5	Online/offline events	Common events
6	In and out of bed incidents	Common events
7	Poor signal event	Common events
8	Anomaly events	Common events
9	Other alarms, the meaning of the alarm is determined by the params parameter: 1-Left the bed 2-Stayed 3-No activity for a long time 4-Sitting alarm 5-Sitting alarm 6-Empty 7-Empty 8-Accessory button alarm 9-Accessory pull rope alarm 10-Accessory voice alarm Note: 6-10 are not open for use	Alerts
10	Entering and exiting the warning area	Common events

- Common events and fall alarms can be obtained without enabling; other alarms require the device to be configured to turn on the corresponding switch to obtain
- It takes more than 2 minutes for the Hobacare platform to detect the power failure or disconnection of the device . To reduce network fluctuations, the Hobacare platform will push the information to the third party through the interface more than 1 minute after it detects the power failure or disconnection. In other words. it will take more than 3 minutes for the power failure or disconnection of the device to be pushed to the third party.

2.3.1. Number of people change event notification format

Parameter name	Type	Required	Value Description	Example
cmd	String	yes	Fixed to " DEVICE_EVENT "	{" cmd ":"DEVICE_EVENT", "count":3, "event":"1",
uid	String	yes	Device UID	

event	St ring	yes	1	"uid":"F59D3E873F5B"}
count	int	yes	Number of people	

2.3.2. Other event notification formats

Parameter name	Type	Required	Value Description
cmd	String	yes	Fixed to " DEVICE_EVENT "
uid	String	yes	Device UID
event	String	yes	Between 2 and 10. For values, please refer to the table at the beginning of Section 2.3.
eventName	String	yes	Event description string
params	Map	no	Except for the fall alarm, all other events have

The different event params parameters are shown in the following table:

Event	Event Type	Params	Example
2	Fall incidents	This parameter is not carried	{" cmd ":"DEVICE_EVENT", "event":"2", " eventName ": "Device detected a fall", "uid":"F59D3E873F5B" }
3	Respiration and heart rate	Breath: Breathing value; heartbeat: heart rate value; alarmType : abnormal type, including: 11: Hyperpnea 12: Hypopnea 13: Apnea 14: Heart rate is too high 15: Heart rate is too low 16: Vital signs are weak	{" cmd ":"DEVICE_EVENT", "event":"3", " eventName ": "The device detected a high heart rate", "params ":{ "breath":40,"heartbeat":110,"alarmType":14}, "uid":"F59D3E873F5B"}
4	In and out door	entry2Exit : " 0 for entry, 1 for exit "	{" cmd ":"DEVICE_EVENT", "event":"4", " eventName ": "User 1 leaves the room", "params ":{ "entry2Exit":"1"}, "uid":"F59D3E873F5B"}
5	Online/Offline	isOnline (0-online 1-offline)	{" cmd ":"DEVICE_EVENT", "event":"5", " eventName ":"Offline", "params ":{ "isOnline":"1"}, "uid":"F59D3E873F5B"}
6	Getting in and out of bed	entry2Exit (0-entry 1-exit)	{" cmd ":"DEVICE_EVENT", "event":"6", " eventName ":"Enter the monitoring bed", "params ":{ "entry2Exit":"0"} ,"uid":"F59D3E873F5B"}
7	Poor signal	recovery (0-trigger 1-recovery)	{" cmd ":"DEVICE_EVENT", "event":"7", " eventName ":"Signal threshold trigger", "params ":{ "recovery":"0"} ,"uid":"F59D3E873F5B"}
8	Abnormal inclination	recovery (0-trigger 1-recovery)	{" cmd ":"DEVICE_EVENT", "event":"8", " eventName ":" Tilt threshold trigger ", "params ":{ "recovery":"0"} ,"uid":"F59D3E873F5B"}
9	Other warnings	alarmType (1- Left out of bed 2- Stayed 3- No one active for a long time 4- Sitting down alarm 5- Sitting up alarm 6- Empty 7- Empty 8- Button SOS alarm 9- Pull rope SOS alarm 10- Call for help	{" cmd ":"DEVICE_EVENT", "event":"9", " eventName ": "Device detected a delay", "uid":"F59D3E873F5B", "params ":{ " alarmType ":"2"} }

		SOS alarm) Note: 6-10 are not open for use	
10	Entering and exiting the warning area	entry2Exit (0-entry 1-exit)	{" cmd ":"DEVICE_EVENT", "event":"10", " eventName ": "Enter the warning area", "params ":{ "entry2Exit":"0"} ,"uid":"F59D3E873F5B"}

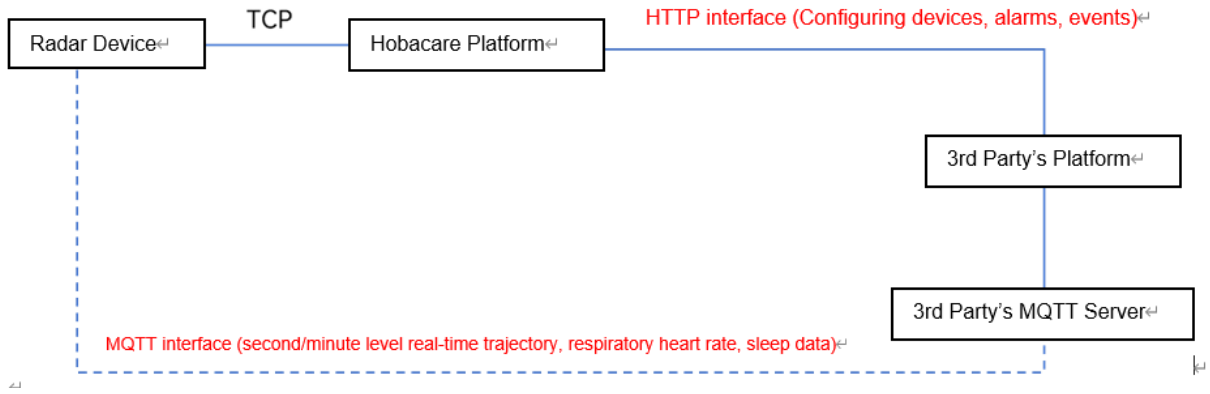
2.3.3. URL callback response

Hobacare platform calls the URL, please return the response in the following format:

Return parameter (**must return the following format**)

Parameter name	Parameter Type	Required	Value Description
code	String/int	Y	200: Success Other: Failed
msg	String	Y	information

3. Quickly connect to real-time data



If the system needs to obtain not only alarm events but also real-time (second-level) trajectory data and respiratory heart rate, you need to deploy an MQTT server. After configuring the MQTT information to the device, the device can upload data in real time.

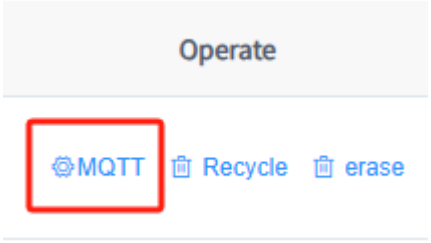
- Only Wi-Fi devices can support uploading. Due to the directional access restrictions of IoT cards , 4G devices need to have a single device order volume of more than 1K to provide corresponding support.

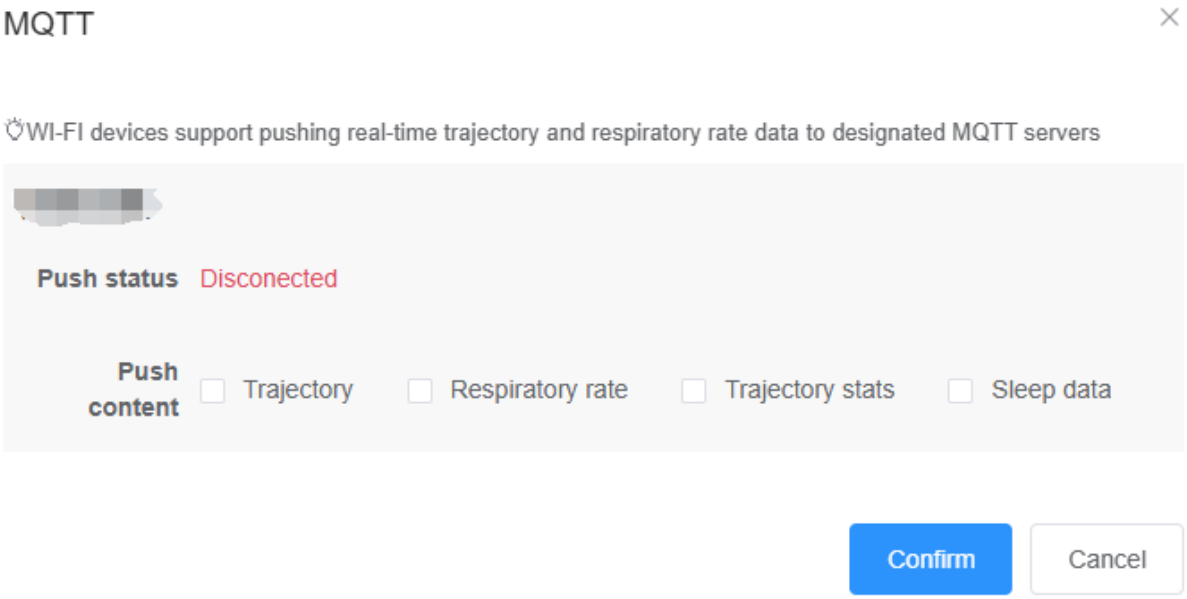
3.1. Deploy MQTT Server

To get real-time data, you need to deploy an MQTT service on your server or use a public cloud MQTT service.
If you have no experience in MQTT service deployment, you can contact Hobacare sales and we can provide paid deployment services.

3.2. Configure device MQTT connection information

After the platform upgrade at the end of October 2024, you will be able to self-configure MQTT connection information in the device list of the Hobacare platform. If automation is required, you can use the API in Section 5.7 to start/close the MQTT connection.
Device management→ Device list





3.3. MQTT upload content format

3.3.1. Release second-level trajectory data

The format of the second-level trajectory message is as follows (you can only focus on the payload part and ignore the header content):

```
{
"header ":{
" traceId ":" xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx ", "payloadVersion":1,
"brand":" qinglan ",
"timestamp":1234567890123, " method":"device-penetrate "
}"payload ":{
"deviceCode":"F59D3E873F8F",
"position":"12D4D5FEE0089"
}
}
```

Where deviceCode is the device UID, position is a base64-encoded byte array of length 16 bytes * N, and N is the number of people detected by the radar device.
When there are people on the vehicle, trajectory data is reported once per second.
When unmanned, the trajectory data is reported once every 30 seconds, and the target ID is fixed at 88.
In the occupied state, each 16 bytes represents a person, which is defined as follows:

Byte	Property Name	Definition
0	Target ID	When someone is in the state, the value is 0-7, representing the person number; When no one is in the state, the value is fixed at 88, indicating that no one is in the state
1~3	Track coordinates	Bytes 1-3 are the x/y/z coordinates respectively. The x and y coordinate units are in decimeters, ranging from -127 to 127; the z coordinate unit is in centimeters, ranging from 0 to 255
4-11	Reserved fields	Reserved for debugging
12	time left	The unit is seconds, the value range is 0-60, and it is only used when automatically measuring the boundary.
13	Personnel posture	0: Initialization 1: Walking 2: Suspected fall 3: Squatting 4: Standing 5: Fall confirmation 6: Lying down 7: Suspected sitting on the ground 8: Confirmed sitting on the ground 9: Sitting up on the ordinary bed 10: Suspected sitting up on the bed 11: Confirmed sitting up on the bed 3.6 The "Posture Display" function needs to be enabled and the firmware needs to be upgraded to version 202406 to support it. 7, 8 need to configure the ground alarm switch to turn on, and need to upgrade to version 202407 or above to support 9-11 requires configuration to turn on the bed sit-up switch, and needs to be upgraded to version 202501 or above to support
14	Personnel events	0: No event 1: Enter room 2: Leave room 3: Enter area 4: Leave area If the device reports an event that is not in the above event definition value, it can be ignored
15	Region ID	When the 14-byte content is 3 or 4, this field identifies the area ID where the person enters and exits

3.3.2. Release second-level breathing and heart rate data

The second-level respiratory heart rate message format is as follows (you can only focus on the payload part and ignore the header content):

{
"header ":{
" traceId ":" xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx ", "payloadVersion":1,
"brand":" qinglan ",
"timestamp":1234567890123, " method":"device-penetrate "
},"payload ":{
"deviceCode":"F59D3E873F8F",
"heartbreath":"12D4D5FEE0089"
}
}

deviceCode is the device UID, and heartbreath is a base64-encoded array of fixed length 16 bytes. When someone enters the monitoring bed, the second-level respiratory and heart rate data are reported at a frequency of once every 2 seconds; otherwise, no data is reported.

The length of the respiratory and heart rate message is fixed at 16 bytes and is defined as follows:

Byte	Property Name	Definition
0	Identifier	Fixed to 0, indicating real-time breathing and heart rate;
1	Breathing value	Unit: times/minute
2	Heart rate value	Unit: times/minute
3-12, 14-15	Reserve	Reserved for debugging
13	Status Events	Bit 7 & bit 6: Represents the sleep state of the past 1 minute 00: Undefined 01: Light sleep 10: Deep sleep 11: Awake

3.3.3. Publish minute-by- minute sleep statistics

the minute-level breathing sleep message is as follows (you can only focus on the payload part and ignore the header content):

{
"header ":{
" traceId ":" xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx ", "payloadVersion":1,
"brand":" qinglan ",
"timestamp":1234567890123, " method":"device-penetrate "
},"payload ":{
"deviceCode":"F59D3E873F8F",
" hbstatics ":"12D4D5FEE0089"
}
}

deviceCode is the device UID, and hbstatics is a base64-encoded array of fixed length 16 bytes. When someone enters the monitoring bed, the minute-level breathing and sleep data are reported once per minute; otherwise, no data is reported.

The message length is fixed at 16 bytes and is defined as follows:

Byte	Property Name	Definition
0	Identifier	Fixed to 0xff, indicating sleep statistics;
1	Real-time breathing value	Unit: times/minute
2	Real-time heart rate value	Unit: times/minute
3-4	Sleep Statistics	The sleep algorithm is not public, and the information in this field is not public.
5	Average respiration per minute	Unit: times/minute
6	Average heart rate per minute	Unit: times/minute
7-12	Sleep Statistics	The sleep algorithm is not public, and the information in this field is not public.
13	Status Events	Status/events that occurred in the past minute Bit 1 & bit 0 represent the breathing status in the past minute: 00: Normal breathing 01: Hypopnea 10: Hyperpnea 11: Apnea Bit 3 & bit 2: represents the heart rate status in the past minute 00: Normal heart rate01: Low heart rate10: High heart rate11: Undefined Bit 5 & bit 4: Represents vital signs 00: Normal status01: Undefined02: Undefined11: Weak vital signs Bit 7 & bit 6: Represents the sleep state of the past 1 minute 00: Undefined 01: Light sleep 10: Deep sleep 11: Awake
14-15	Sleep Statistics	The sleep algorithm is not public, and the information in this field is not public.

Note that except for weak abnormalities in vital signs, it is generally not recommended to immediately generate an alarm based on minute-level respiratory and heart rate events.

In addition, the sleep status in the minute-level data is calculated in real time, which will be slightly different from the sleep status calculated over a long period of time in the daily sleep report.

3.3.4. Publish minute-level trajectory statistics

the minute-level trajectory statistics message is as follows (you can only focus on the payload part and ignore the header content):

{
"header ":{
" traceId ":" xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx ", "payloadVersion":1,
"brand":" qinglan ",
"timestamp":1234567890123, " method":"device-penetrate "
},"payload ":{
"deviceCode":"F59D3E873F8F",
"posstatics":"12D4D5FEE0089"

}
}
Where deviceCode is the device UID, and posstatics is a base64-encoded array of fixed length 16 bytes.
Regardless of whether there is someone present, this data is reported at a fixed frequency of once per minute;
the minute-level statistics message is fixed at 16 bytes and is defined as follows:

Byte	Property Name	Definition
0	Version Identifier	The current value is 1 or 2
1	Number of people	The number of people detected by the radar at the current reporting time. Value range: 0-8
2~3	Walking distance	The walking distance counted in the past minute, in meters. The value range is 0-32767 and is filled in Big-Endian byte order. Note: No statistics are taken for multiple people.
4	Walking time	The walking time in the past minute, in seconds Note: No statistics are taken for multiple people.
5	Meditation duration	sitting time in the past minute , in seconds Note: No statistics will be made in case of multiple people . Sit still will only be recognized within 3*3 meters of the top installation , and will not be recognized beyond this range.
6	Length of time in bed	in the monitoring bed in the past minute , in seconds Note: No statistics are taken for multiple people.
7	Standing time	The static time in the past minute, in seconds Note: No statistics are taken for multiple people.
8	Multiplayer duration	The time that multiple people have been present in the past minute, in seconds During this period, no statistics will be collected on walking distance, walking time, or stationary time.
9	Logo	Used when the value of byte 0 is >= 2, otherwise it cannot be used Bit 0: Indicates whether the breathing and heart rate measurement is currently being performed. The other bits are not used yet
10 ~15	rsv	Not open for use

4. Web/ WeChat Mini Program/APP Development

4.1. Rapid Web Development

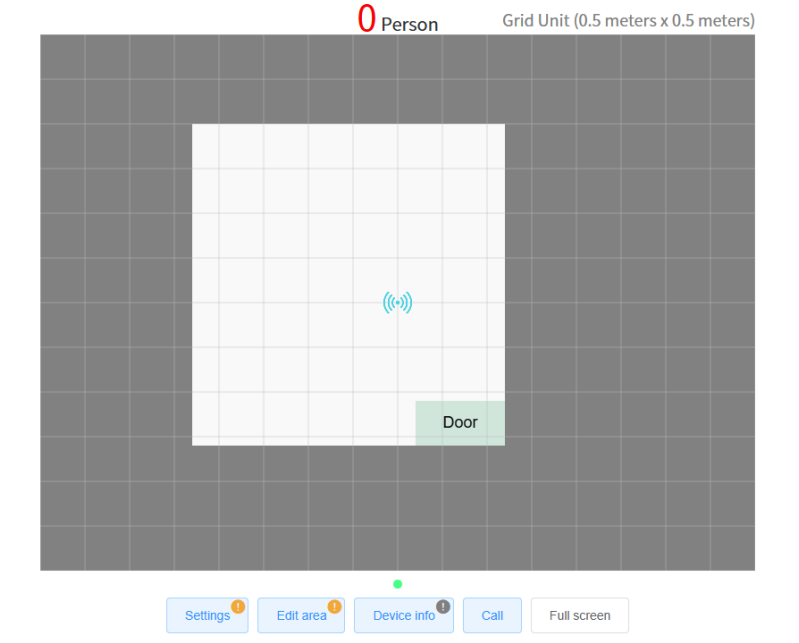
If you want to reuse some web pages of the Hobacare platform, you can use iframe to embed the page and quickly integrate it into your system.
If the web page of the Hobacare platform does not meet your requirements, you will need to refer to the interface in Chapter 5 and the events and real-time data obtained in Chapters 2/3 for new development.

4.1.1. Web Embedded Authentication

Before reusing the HobacareWeb page, you need to obtain an authentication token. Please read sections 5.1 and 5.6.1 to ensure that you can notify the API interface to obtain the token value required for Web embedding.

4.1.2. Device details page

The device details page provides a device's real-time trajectory, breathing and heart rate, function attribute configuration, area configuration, device profile configuration, sleep report and digital portrait, and event center content.



Access path and parameter description

https://qinglanst.com/single/rommDetails?uid=xxxx&token=xxxx&freshToken=xxxx&readOnly=x_xxx_&_lang_=xxx_x

property	type	Required	illustrate
uid	String	Y	Device unique uid

token	String	Y	Refer to 4.1.1 to obtain
freshToken	String	Y	
readOnly	Boolean	N	The default value is false. If set to true, the edit area/function attribute/device profile button will not be clickable
lang	String	N	The default is zh_CN , you can set zh_CN , en_US , ja_JP , th_TH

4.1.3. Call Device Page

Call Device page allows you to call HC2 Wi-Fi devices that support Wi-Fi calling (Note: 4G devices only support direct dialing of numbers by landline or mobile phone, not web page calls).



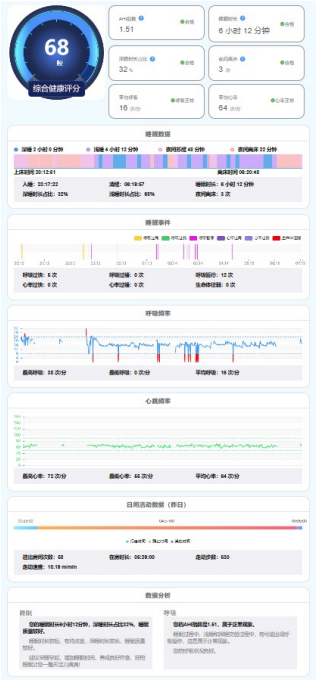
Access path and parameter description

https://qinglanst.com/metting?uid=xxxx&token=xxxx&freshToken=xxxx_&_lang_=xxx_x

Property	Type	Required	Illustrate
uid	String	Y	Device unique uid
token	String	Y	Refer to 4.1.1 to obtain
freshToken	String	Y	
lang	String	N	The default is zh_CN , you can set zh_CN , en_US , ja_JP , th_TH

4.1.4. Sleep report/digital portrait page

The sleep report and digital portrait pages can be opened by clicking a button in the embedded page of 4.1.2, or they can be opened directly in the following way.





Access path and parameter description

https://qinglanst.com/single/rommDetails?uid=xxxx&token=xxxx&freshToken=xxxx&defaultExpand=xxx_&_lang_=xxx_x

property	type	Required	illustrate
uid	String	Y	Device unique uid
token	String	Y	Refer to 4.1.1 to obtain
freshToken	String	Y	
defaultExpand	String	Y	Filling in "sleepReport" will open the sleep report directly Fill in "digitalImage" to open the digital image directly
lang	String	N	The default is zh_CN , you can set zh_CN , en_US , ja_JP , th_TH

4.1.5. Large screen page

4.1.5.1. Large screen home page



Access path and parameter description

https://qinglanst.com/daping?token=xxxx&freshToken=xxxx_&_lang_=xxx_x

Property	Type	Required	Illustrate
token	String	Y	Refer to 4.1.1 to obtain
freshToken	String	Y	
lang	String	N	The default is zh_CN , you can set zh_CN , en_US , ja_JP , th_TH

4.1.5.2. Large screen device list



Access path and parameter description

https://qinglanst.com/daping-devList?token=xxxx&freshToken=xxxx_&_lang_=xxx_x

Property	Type	Required	Illustrate
token	String	Y	Refer to 4.1.1 to obtain
freshToken	String	Y	
lang	String	N	The default is zh_CN , you can set zh_CN , en_US ,

			ja_JP , th_TH

4.1.5.3. Large screen device details page



Access path and parameter description

https://qinglanst.com/daping-detail?uid=xxxx&token=xxxx&freshToken=xxxx_&_lang_=xxx_x

Property	Type	Required	Illustrate
uid	String	Y	Device unique uid
token	String	Y	Refer to 4.1.1 to obtain
freshToken	String	Y	
lang	String	N	The default is zh_CN , you can set zh_CN , en_US , ja_JP , th_TH

4.2. Mini Program/APP Rapid Development

Hobacare platform provides network distribution SDK/network distribution jump applet and device management H5 page for embedding.

4.2.1. Wi-Fi device network configuration

Hobacare provides APP network configuration SDK for development:

Android source code: https://github.com/EspressifApp/EspBlufi

IOS source code: https://github.com/EspressifApp/EspBlufiForiOS

If you use the mini program, you can jump to the configuration page of the Hobacare whiteboard mini program to configure the network. No authentication is required when jumping.

This page connects the device by scanning a QR code or Bluetooth , and then configures the device's Wi-Fi connection username and password.

If you need to change the server address, click "Advanced" on the network configuration page and select "Server Address" to change it.



Configure the web page jump path and parameters:

```
wx.navigateToMiniProgram ({
  appId : 'wx9a07688c72704f57'
path: ' pgkSub /device-type/ index?isOnlyNetwork = true&lang = xxxx '
})
```

Parameters:

@param {String} lang zh_CN , en_US Optional

4.2.2. Phone Call Page

As with the Web, the page can be adapted to the phone's resolution. See 4.1.3.

4.2.3. Device details page

The device details page can provide trajectory, breathing and heart rate display, and provides functions such as functional attribute setting, boundary area setting, device profile setting, event center viewing, sleep report viewing, and device calling.

The system provides two ways of docking: H5 embedding and mini-program jump.



H5 access path:
https://qinglanst.com/h5/#/pages/home/index?uid=xxxx&url_token=xxxx&lang=xxxx
Parameter:
@param {String} uid //Device UID, Mandatory
@param {String} url_token //Refer to section 5.6.2, Mandatory

@param {String} lang zh_CN、en_US //Optional

4.2.4. Sleep Report (H5)

The sleep report used by the mobile version is obtained through the following access path and parameters



Access path:
https://qinglanst.com/h5/#/pgkSub/sleep-report/index?uid=xxxx&date=xxxx&url_token=xxxx&lang=xxxx

Parameter:
@param {String} uid //Device UID, Mandatory
@param {String} date //Date, Mandatory
@param {String} url_token //Refer to section 5.6.2, Mandatory @param {String} lang zh_CN、en_US //Optional

5. Advanced Features: HTTP API

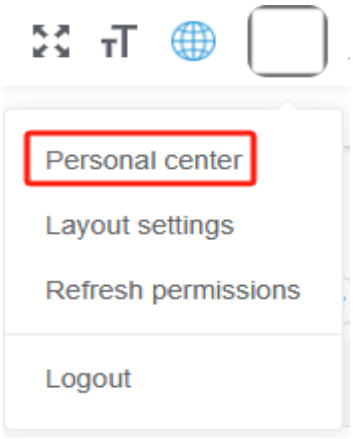
If you need to define a new UI interface to configure and manage Hobacare devices, you will need to use the HTTP API mentioned in this chapter.

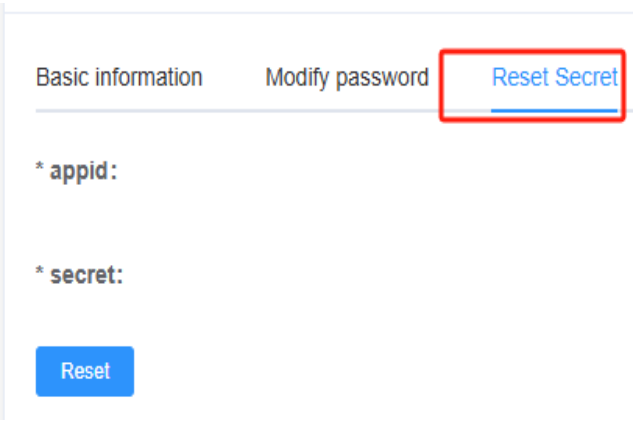
5.1. API Authentication Rules

5.1.1. Get appid and secret

Use the administrator account to log in to the Hobacare platform https://qinglanst.com/
Click the avatar in the upper right corner, select "Personal Center", find the "Reset Key" page in the basic information, and click Reset to obtain the key for docking.

- Please remember the key information. You will need the key to access the subsequent docking interface.
- If you forget the key, the system cannot be retrieved and can only be reset and regenerated





5.1.2. HTTP interface authentication rules

All HTTP request messages sent to the Hobacare platform need to add the following three fields in the Request Header for authentication (key is case-sensitive):

Key	Value
appid	Refer to Section 5.1.1 to obtain
timestamp	UNIX timestamp, which is the number of seconds since 00:00:00 on January 1, 1970
signature	The signature value generation rule is: **SHA-1** (secret # timestamp # Data) **secret** : refer to section 5.1.1 to obtain **timestamp** : UNIX timestamp, see the previous line of the table **Data: Sort** the content of the HTTP request in ***lexicographical order by key*** and concatenate them using the key=value method Note: The current HTTP request content has two forms: Query and Body. Regardless of the form, it is spliced according to the above rules. For example, {"a":"a","c":"c"," map ":{"e":"e"} , " list " : " [2,1,3] "} => a=a #c=c #list= 2=1=3#map=e=e #

Full example:

Example 1 (Subscription interface):

the appid obtained from Section 5.1.1 is ql320230911123015310a, the secret is e6accd9ccabac94453c91f29d8638253d0f36a1a, the current timestamp is 1723112313 , and the data transmitted by the interface is {"eventUrl":"http://test.xxx.com/test","event":["1","2"],"humanUrl":"http://test.xxx.com/test"}

First, concatenate and sort the above data to get the string e6accd9ccabac94453c91f29d8638253d0f36a1a#1723112313#event=1=2#eventUrl=http://test.xxx.com/test#humanUrl=http://test.xxx.com/test#

- Please note that you need to add " # " at the end when splicing.

Calculate the SHA-1 value of the concatenated string
SHA-1 (the concatenation of the above) = 67980ABD4F1031B8299E0772E263C1D7999D68B2

The HTTP request message header sent to Hobacare needs to have the following content:

Key	Value
appid	ql320230911123015310a
timestamp	1723112313
signature	67980ABD4F1031B8299E0772E263C1D7999D68B2

Example 2 (open the mqtt interface):

the appid obtained from Section 5.1.1 is ql320230911123015310a, the secret is e6accd9ccabac94453c91f29d8638253d0f36a1a, the current timestamp is 1723112030 , and the data transmitted by the interface is {"uid":"9D8A32CEBA1B","server":"192.168.1.188","password":"sun123","clientId":"2024-06-13","keepAlive":60,"messageType":["1","2"],"port":1883,"topics":{"pub":["/topic/pub/sun/device"]},"username":"sun_test"}

First, concatenate and sort the above data to get the string e6accd9ccabac94453c91f29d8638253d0f36a1a#1723112030#clientId=2024-06-13#keepAlive=60#messageType=1=2#password=sun123#port=1883#server=192.168.1.188#topics=pub=/topic/pub/sun/device#uid=9D8A32CEBA1B#username=sun_test#

- Please note that you need to add " # " at the end when splicing.

Calculate the SHA-1 value of the concatenated string
SHA-1 (the concatenation of the above) = FF0B66D038E986B951AA880A8949341649BE592D

The HTTP request message header sent to Hobacare needs to have the following content:

Key	Value
appid	ql320230911123015310a
timestamp	1711597995
signature	FF0B66D038E986B951AA880A8949341649BE592D

Attached is a case study

5.1.3. English support

For the sleep report interface of 5.5.1, if you need to use English, you need to add the following content to the HTTP request header :
' Cookie ' : ' language = en_US '

5.2. Device Binding Delete API

5.2.1. getDeviceInfo (Get device list)

Functional Description

Get all devices and online status under the enterprise account.
It is recommended to call it once when the system starts or restarts to achieve data synchronization, and then synchronize the status by subscribing to device events.

Interface restrictions

⚠ The Hobacare platform limits a connected IP to calling this interface twice per minute. Avoid frequent calls.

Request method

HTTP GET application/x-www-form- urlencoded

Request address

https://qinglanst.com/ prod-api/thirdparty/v2/getDeviceInfo

Request Parameters

none

Return to content

Property	Type	Required	Illustrate
code	String	Y	200 - Success 500 - Failure
msg	String	Y	
data	Array<device>	Y	Device object array, the device object structure is shown below

Device object structure:

Parameter name	Type	Required	Illustrate
uid	String	Y	Device UID
e qt _n ame	String	Y	Device custom name. Default is empty.
isOnline	String	Y	Whether online. 0 means online, 1 means offline
modelNumber	number	Y	Device Model 1-HC1 2-HC2 3-HC2N 4-HC2F 5-HC2S 6-HC2-4G 7-HC2F-4G

			8-HC2S-4G 9-TK2 10-TK2N 11-TK2F 12-TK2S 13-HC2N-4G

Example

```
Return to content
{
      " code " : " 20 0" ,
       " msg " : " success " ,
       " data " : [ {
      " uid ": "75BA300247D3",
      " eqt  Name  " : "60G radar",
      " isOnline ": "1" ,
      " modelNumber ": 1 ,
      } ]
}
```

5.2.2. bindDevice (bind device)

Functional Description

Through the device UID, it will be bound to the enterprise account.

Interface restrictions

⚠ If the bound device does not exist or an attempt is made to bind a device belonging to another company, an illegal operation prompt will be generated. If there are more than 5 illegal operation prompts within 30 minutes, the interface will be locked for 20 minutes.

Request method

HTTP POST application / json format

Request address

https://qinglanst.com/ prod-api/thirdparty /v2 /bindDevice

Request Parameters

Parameter name	Type	Required	Value Description
uid	String	Y	Device UID

Return to content

Property	Type	Required	Illustrate		
code	String	Y	" 200 " indicates success "712" indicates illegal operation "711" means the device has been bound		
msg	String	Y			
data	Obj	N	Code-200 returned		
			isOnline	string	0 means online, 1 means offline
			modelNumber	number	1-HC1 2-HC2 3-HC2N 4-HC2F 5-HC2S 6-HC2-4G 7-HC2F-4G 8-HC2S-4G 9-TK2 10-TK2N 11-TK2F 12-TK2S 13-HC2N-4G

5.2.3. unbindDevice (Unbind Device)

Functional Description

Unbind the device from this enterprise. After the device is unbound, it will be in an unbound state and can be bound to other enterprises for use.

Interface restrictions

none

Request method

HTTP POST application / json format

Request address

https://qinglanst.com/ prod-api/thirdparty /v2 /unbindDevice

Request Parameters

Parameter name	Type	Required	Value Description
uid	String	Y	Device UID

Return to content

Property	Type	Required	Illustrate
code	String	Y	" 200 " means success, and push will stop automatically after unbinding "500" means the device does not exist or the current device is assigned to a Hobacare Technology Mini Program user. The Mini Program user needs to unbind the device before the backend can unbind the device from the enterprise.
msg	String	Y	

5.2.4. subscribeAffair (Subscribe to device events)

Functional Description

Register a callback URL and subscribe to events of all devices in the enterprise on demand. Note: This function can also be configured through the web interface described in Section 2.2.

Interface restrictions

⚠If you call this API multiple times, the last call will prevail.

⚠It is recommended that the interface be processed in multiple threads and return a success message immediately after receiving the message.

⚠ After calling this API, the stock interface is https://qinglanst.com/prod-api/thirdparty/v2/subAffair The result is invalid, please refer to the result of this API.

Request method

HTTP POST application / json format

Request address

https://qinglanst.com/prod-api /thirdparty/v2/subOwner

Request Parameters

Parameter name	Type	Required	Value Description
event	Array <String>	yes	If you only need to subscribe to some events, you can specify an array of event IDs. See the next table for a list of event IDs.
humanUrl	String	yes	Number of people callback URL, if not, it is the same as eventUrl
eventUrl	String	yes	Event callback URL, if not specified, it will be the same as humanUrl

The event IDs and generation conditions supported by the device are as follows:

Event ID	Event Name	Conditions	Restrictions
1	Changes in the number of people	When the number of people detected by the radar changes, an event is generated to notify the latest number of people. If the number of people remains unchanged, it will be pushed every 15 minutes	Firmware version: February 2024 or later
2	Fall incidents	When the radar detects a fall event	Equipment model: HC2, TK2, HC2F, TK2F Working mode: fall detection or bed monitoring
3	Respiration and heart rate	This occurs when the radar detects that the breathing rate is higher or lower than the threshold. For threshold settings, refer to 5.3.5	Equipment model: HC2, TK2, HC2S Working mode: breathing, sleeping or bed monitoring
4	In and out door	When the radar detects a person entering or leaving the door	none
5	Online and offline	Generated when the device is online or offline. Note: After the device is powered on or off, it takes about 1 minute for the platform to sense and report the event.	none
6	Getting in and out of bed	When the radar detects a person entering or leaving the bed	none
7	Poor signal	Generated when the radar network signal exceeds the normal threshold range. Wifi normal range: -88~-20 4g Normal range: 11~31	Firmware version: March 2024 or later
8	Abnormal inclination	Generated when the radar dip falls below the threshold. When top-mounted , the tilt angle should not exceed 10° When side mounted, the wall angle should be within 30° Note: Only the HC2 series supports inclination measurement, the TK2 series does not support	Firmware version: March 2024 or later
9	Other warning events	1-Leaving the bed and not returning (the time spent away from the bed reaches a threshold within a certain period of time) 2-Detention (triggered when the time in the room reaches the threshold) 3-Long-term no-activity (24-hour no-activity, check every morning) 4-Sitting alarm (person falls down in a sitting position within ± 1.5*±1.5 meters below the radar and reaches the time threshold) 5-Bed sit-up alarm (person stands up after lying in bed) 6-Empty 7-Empty 8-Push button SOS alarm 9-Pull rope SOS alarm 10-Call for help SOS alarm (reserved for future accessories, not open)	Working mode: fall detection, breathing sleep or bed monitoring with interface setDeviceProp to set parameters
10	Entering and exiting the warning area	7-When the radar detects a person entering or leaving the warning area,	none

Return to content

Property	Type	Required	Illustrate
code	String	Y	200 - Success 500 - Failure
msg	String	Y	

5.3. Device Configuration API

5.3.1. getDeviceProp (Get device properties)

Functional Description

the function attributes of the device through the device UID

Interface restrictions

⚠Device properties are stored in the device's flash and are available only when the device is online.

Request method

HTTP GET application/x-www-form- urlencoded

Request address

https://qinglanst.com/prod-api/thirdparty/v2/deviceProp

Request Parameters

Parameter name	type	Required	Value Description
uid	String	Y	Device UID

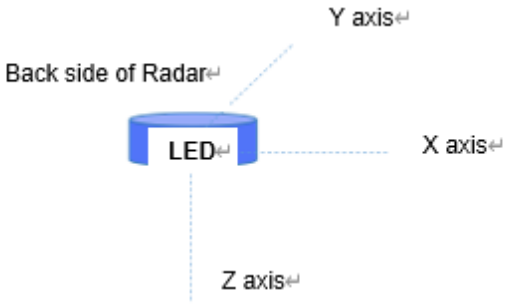
Return to content

Property	Type	Required	Illustrate
code	String	Y	" 200 " indicates success " 500 " means failure " 777 " means offline
msg	String	Y	
data	Map	Y	See the table below

Details of data

Property	Type	Required	Illustrate
radar_func_ctrl	String	Y	Working Mode: 15 - Bed monitoring 11 - Breathing and sleeping 7 - Fall monitoring 3 - People tracking
radar_install_style	String	Y	Installation method, for values, refer to 5.3.2 description
suspected_fall_time	String	Y	Fall alarm time, in 10 seconds, same as postureParams If there is a conflict between the parameters, use only one of them. For the value, refer to the description in 5.3.2
radar_install_height	String	Y	Installation height, in decimeters. For values, refer to

			5.3.2.
r ectangle	String	Y	Equipment installation boundary. Returned in the form of four vertex coordinates { lowLeft ; lowRight ; upLeft ; upRight }. For the meaning of the four vertices, refer to 5.3.3. Assuming that in the side-mounted case, the positive direction of the X axis is 2 meters, the negative direction of the X axis is 2.2 meters, and the positive direction of the Y axis is 4 meters, the returned result is {-22,0;20,0;-22,40;20,40}
declare_area	String	Y	Locale. Returned in the form of { coordKey , type, lowLeft , lowRight , upLeft , upRight }. Refer to 5.3.4 for the meaning of these values. that area 0 is currently set as a door and area 1 is a monitoring bed, the returned results are as follows: {0, 4,- 15,20,15,20,-15,25,15,25},{1,5,-6,0,6,0,-6,20,6,20},
heart_breath_param	String	Y	Breathing and heart rate parameters. The return format is an array, in which only the first 7 numbers are valid. [upperBreathe , upperHeartRate , lowerBreathe , lowerHeartRate , ...] For the meaning of the numbers, see the setBreathHeartParam interface. if the respiration high threshold is 32, the heart rate high threshold is 100, the respiration low threshold is 5, the heart rate low threshold is 50, continuous measurement is on, the vital signs weak trigger is 10 minutes, and the sensitivity is 5, the following content is returned (only the first 6 values are shown): [32,100,5,50, 1 , 10 , 5,0,0,0 , 0,0,0,0,0, 0] Tip: Since the value is stored in one byte, a value exceeding 127 will become a negative number, and you need to use &0xFF to get the original value. For example, -126&0xFF = 130
signal_intensity	String	Y	The device signal strength value, the larger the better. Wifi : < = -100 No signal (-100, -88] Difference (-88, -66] General (-66, -55] Not bad >= -55 Strong 4G - A strong signal generally refers to a CSQ value between 23 and 31. - A medium signal has a CSQ value between 15 and 22. - The CSQ value of weak signals is between 0 and 1. Wi-Fi is available ' - ' 4G is not available '-'
accelera	String	Y	Radar inclination. The format is X:Y:Z:V four values. V represents whether the radar angle has been factory calibrated, 0 represents uncalibrated, and 1 represents calibrated. XYZ represents the angle between the radar and the three axes of space XYZ. In the case of ceiling

			mounting , X and Y should be within ±10°; in the case of side mounting, X should be within ±10°, and Y should be between -60° and -90° (within 30° of the wall angle). 
radar_compile_time	String	Y	Radar compilation time, currently used to determine the device version
app_compile_time	String	Y	Communication compilation time, currently used to determine the device version
leaveAlarmSwitch	String	N	Bed leaving alarm switch 0-on 1-off
leaveDetectionTime	Number	N	Bed leaving alarm triggering time (unit: minutes)
leaveDetectionRange	String	N	The format of the time interval for triggering the bed leaving alarm is HH:mm-HH:mm Example: 08:00-20:00 or 20:00-08:00 Leaving the bed during this interval will trigger
longAwaySwitch	String	N	Long-time absence monitoring switch 0-on 1-off
detentionAlarmSwitch	String	N	Detention alarm switch 0-on 1-off
entryDetectionTime	Number	N	Detention alarm triggering time unit: minutes
suddenDeathSwitch	String	N	Vital sign weak alarm switch 0-on 1-off
postureParams	String	Y	Posture parameters (English, separated) For example: 2 , 3 , 5Analysis : First parameter: The suspected fall determination time is set in units of 10 seconds. Minimum setting: 1*10 seconds Maximum setting: 10*30 seconds = 5 minutes Setting a value of 0 will use the default value of 3 (i.e. 30 seconds) instead. with suspected_fall_time, choose one Second parameter: bit0: Whether the sitting alarm is turned on bit1: Whether the posture detection is turned on

			(bit0 or bit1 is enabled This bit is automatically enabled) bit2 : Whether the sitting up alarm in bed is turned on We will continue to expand other bits to do other functions in the future The third parameter: The sitting alarm judgment time is set in units of 10 seconds. Minimum setting: 1*10 seconds Maximum setting: 10*30 seconds = 5 minutes Setting a value of 0 will use the default value of 3 (i.e. 30 seconds) instead.

Example

{
"msg": " Operation successful ",
"code": 200,
"data": {
" radar _install_style ": "1",
" accelera ": "-1. 06:- 62.89:-62.92:1",
" heart _breath_param ": "[40, 100, 8, 50, 0, 11, 5, 10, 0, 0, 0, 0, 0, 0, 0, 0]",
" postureParams ": "2,3,5",
" signal _intensity ": "-44",
" suspected _fall_time ": "2",
" leaveDetectionRange ": "08:00-20:00",
" radar _func_ctrl ": "15",
" leaveAlarmSwitch ": "0",
" radar _install_height ": "18",
" longAwaySwitch ": "1",
" suddenDeathSwitch ": "1",
" app _compile_time ": "Nov 26 2024 16:20:51",
" entryDetectionTime ": 0,
"rectangle": "{-30,0;11, 0;- 30,30;11,30}",
" detentionAlarmSwitch ": "1",
" radar_compile_time ": " Dec 4 2024 08:53:22",
" declare_area ": "{0, 5,- 10,0,5,0,-10,15,5,15},{1,4,10,0,15,0,10,30,15,30},{2,4,-30,0, -25,0,-30,30,-25,30},{3,3,-15,20,0,20,-15,30,0,30},",
" leaveDetectionTime ": 15
}
}

5.3.2. setDeviceProp (Set basic properties)

Functional Description

Set the device's working mode, installation method, installation height, and fall parameters through the device UID.

Interface restrictions

⚠Device properties are stored in the device's flash and can only be set when the device is online.

Request method

HTTP POST application/ json

Request address

https://qinglanst.com/prod-api/thirdparty/v2/prop

Request Parameters

Parameter name	Type	Required	Value Description
uid	String	Y	Device UID
scene	String	N	Working mode. The value range is 1-3, representing: 1-Bed monitoring 2-Fall monitoring 3-People tracking 4-Respiratory sleep (firmware support from March 2024) The default value for HC2/TK2 is 1, the default value for HC2N/TK2N is 3, the default value for HC2F/TK2F is 2, and the default value for HC2S is 4
installType	String	N	Installation mode. The value is 0 or 1. The default value is 0, which means top installation. 1 means side installation.
height	Number	N	Installation height. The unit is decimeter. The maximum setting range is 15-33. The default value is 28, which means the height is 2.8 meters.
susFallDuration	Number	N	Suspected time of fall. The unit is 10 seconds, and the maximum setting range is 0-30. The default value is 6, which means 60 seconds.
leaveAlarmSwitch	String	N	Bed leaving alarm switch 0-on 1-off
leaveDetectionTime	Number	N	Bed leaving alarm triggering time (unit: minutes)
leaveDetectionRange	String	N	The format of the time interval for triggering the bed leaving alarm is HH:mm-HH:mm Example: 08:00-20:00 or 20:00-08:00 Leaving the bed during this interval will trigger
longAwaySwitch	String	N	Long-time absence monitoring switch 0-on 1-off
detentionAlarmSwitch	String	N	Detention alarm switch 0-on 1-off
entryDetectionTime	Number	N	Detention alarm triggering time unit: minutes
suddenDeathSwitch	String	N	Vital sign weak alarm switch 0-on 1-off
postureParams	String	N	Posture parameters (English, separated) For example: 2 , 3 , 5Analysis

			: First parameter: The suspected fall determination time is set in units of 10 seconds. Minimum setting: 0 seconds Maximum setting: 10*30 seconds = 5 minutes The second parameter: bit0: Whether the sitting alarm is turned on bit1: Whether the posture detection is turned on (bit0 or bit1 is enabled This bit is automatically enabled) bit2 : Whether the sitting up alarm in bed is turned on We will continue to expand other bits to do other functions in the future The third parameter: The judgment time of the sitting alarm is set in units of 10 seconds. Minimum setting: 1*10 seconds Maximum setting: 10*30 seconds = 5 minutes Setting a value of 0 will use the default value of 10 (i.e. 100 seconds) instead.

⚠ 1. The installation method and installation height must be set at the same time, otherwise it will not take effect
2. The three parameters of the bed leaving alarm must be set at the same time, otherwise it will not take effect
3. Both parameters of the retention alarm must be set at the same time, otherwise they will not take effect
4. The bed leaving alarm, detention alarm, and long-time absence monitoring switch need to be passed in together with the scene parameter, otherwise they will not take effect.

- When scene is 3 or 4, no fall detection is performed. The susFallDuration parameter can be set but will not take effect.
- When scene is 4, the installation method and height can be set but will not work. The radar firmware will use the side-mounted height of 1.5 meters.
- If you switch installation methods, the margins will revert to the default values.

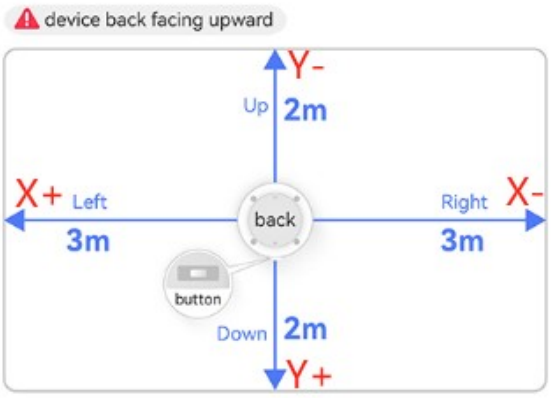
Return to content

Property	Type	Required	Illustrate
code	String	Y	" 200 " indicates success " 500 " means failure " 777 " means offline "778" means that the device is not suitable for this mode
msg	String	Y	

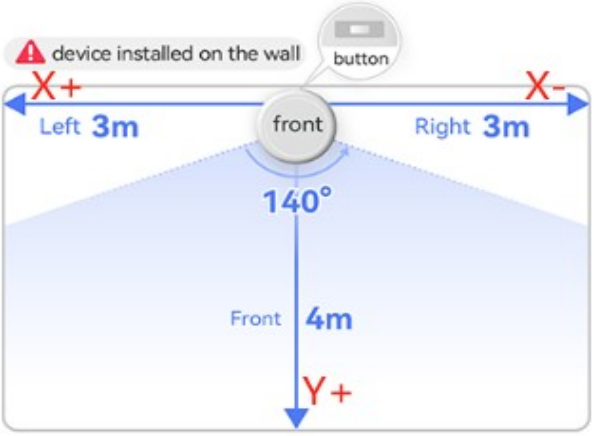
5.3.3. setDeviceBoundary (Set Installation Boundary)

Functional Description

Through the device UID, set the coordinates of the four vertices of the device's detection boundary to determine the boundary range.
The radar is divided into X-axis and Y-axis, and distinguishes between positive and negative axes. Depending on the top-mounted or side- mounted situation, the XY axes and positive and negative signs are shown in the figure below. When setting the boundary, you need to configure the coordinates of the four vertices, and their names and positions are shown in the figure:



Top-mounted directional diagram



Side- mounted directional diagram

By default, the four vertices of the detection boundary are shown in the following table (note that the coordinate values are in decimeters):

Installation	Vertex Name	Default coordinate values	Illustrate
Top loading X axis ±3 meters Y axis ±2 meters	lowLeft	- 30, - 20	lowLeft is in the negative direction of the X axis and the negative direction of the Y axis, and the XY values are negative
	lowRight	30, -20	lowRight is in the positive direction of the X axis and the negative direction of the Y axis, the X value is positive and the Y value is negative
	upLeft	-30, 20	upLeft is in the negative direction of the X axis and the positive direction of the Y axis. The X value is negative and the Y value is positive.
	upRight	30,20	upRight is in the positive direction of the X axis and the positive direction of the Y axis, and the XY values are positive
Side Mount X axis ±3 meters Y axis 4 meters	lowLeft	-30,0	on the side-mounted Y axis , the minimum is 0. The rest has the same meaning as the top-mounted
	lowRight	30, 0	
	upLeft	-30, 40	
	upRight	30,40	

Interface restrictions

⚠Device properties are stored in the device's flash and can only be set when the device is online.

Request method

HTTP POST application/ json

Request address

https://qinglanst.com/prod-api/thirdparty/v2/border

Request Parameters

Parameter name	Type	Required	Value Description
uid	String	yes	Device UID
upLeft	String	yes	x,y format. Please refer to the function description for its meaning. Be careful not to make mistakes with the vertex position and numerical sign. Note: If the four values are set to -127, the radar will enter the automatic measurement boundary; if set to 127, the automatic measurement boundary will stop. If it does not stop for 1 minute after entering the automatic measurement boundary, the radar will automatically stop and restore the boundary to the default value.
lowLeft	String	yes	
upRight	String	yes	
lowRight	String	yes	

Return to content

Property	Type	Required	Illustrate
code	String	Y	" 200 " indicates success " 500 " means failure " 777 " means offline
msg	String	Y	

Example

ask

```
{
"uid":"F59D3E873F5B",
" lowLeft ":" - 30 , - 20 ",
"upLeft":" 30 , - 20 ",
"upRight":" 30 , 20 ",
"lowRight":" 30 , - 20 "
}
```

Return Parameter

```
{
      " code " : " 20 0" ,
      " msg " : " success " ,
}
```

5.3.4. setDeviceArea (Set area)

Functional Description

Set the device area information (including door, bed, interference area, and custom area) through the device UID.

Interface restrictions

⚠Device properties are stored in the device's flash and can only be set when the device is online.

Request method

HTTP POST application/ json

Request address

https://qinglanst.com/prod-api/thirdparty/v2/coords

Request Parameters

Parameter name	Type	Required	Value Description
uid	String	yes	Device UID
coordKey	String	yes	Area number, value range is 0-15. The system can set up to 16 areas. When calling this interface to set the same area number multiple times, the last setting will prevail.
type	Byte	yes	0x00 //Invalid area, indicating that this area is set to invalid (delete the area) 0x01 //Custom area, used to identify objects, no events are generated 0x02 //Bed area 0x03 //Interference area, prohibit any alarm events 0x04 //Door area, in actual use, take the median of the four coordinates 0x05 //Monitoring bed area (refer to the following notes) 0x06 //Warning area. Entering this area will generate an event notification. Note: After the 0x02 type bed area is set, if it meets the monitoring bed range requirements, the radar device will change its type to 0x05, indicating that it is a bed with respiratory heart rate and sleep detection capabilities.
upLeft	String	no	The coordinates of the region vertices. For coordinate definition, refer to 5.3.3
lowLeft	String	no	
upRight	String	no	
lowRight	String	no	

Return to content

Property	Type	Required	Illustrate
code	String	Y	" 200 " indicates success " 500 " means failure

			" 777 " means offline
msg	String	Y	

5.3.5. setBreathHeartParam (Set breathing heart rate parameters)

Functional Description

Adjust the parameters of the device's high/low respiratory and heart rate events through the device UID.
When the user enters the monitoring bed and lies down in single-person mode, and measures the breathing and heart rate stably for one minute, an event is generated if the breathing and heart rate meet the following conditions:

Classification	event	Conditions
breathe	Too high	Exceeding the threshold for 1 minute
	Too low	Below the threshold for 1 minute
	pause	No breathing signal can be detected for about 20-30 seconds within 1 minute
Heartbeat	Too high	Exceeding the threshold for 1 minute
	Too low	Below the threshold for 1 minute
	Weak signs	No heartbeat signal can be measured for 10 consecutive minutes

Note: For apnea events, less than 5 times within 1 hour is considered normal in medicine and should be used as long-term observation data. It is not appropriate to trigger an immediate telephone alarm.

Interface restrictions

⚠Device properties are stored in the device's flash and can only be set when the device is online.

Request method

HTTP POST application/ json

Request address

https://qinglanst.com/prod-api/thirdparty/v2/threshold

Request Parameters

Parameter name	Type	Required	Value Description
uid	String	yes	Device UID
upperBreathe	number	yes	Hyperpnea threshold. Setting range 0-255. Default value is 24.
upperHeartRate	number	yes	Heart rate high threshold. Setting range 0-255. Default value is 90.
lowerBreathe	number	yes	Hypopnea threshold. Setting range 0-255. Default value is 8.
lowerHeartRate	number	yes	Heart rate low threshold. Setting range 0-255. Default value is 50.

intensiveCare	number	no	Continuous measurement switch 0 is off 1 is on Default 0 Not transmitted Use the default value to overwrite Restriction: Must be in breathing sleep mode
suddenDuration	number	no	The unit of the weak trigger time of vital signs is minutes. The default value is 10. If not transmitted, the default value will be used.
sensitivity	Number	no	Sensitivity 1-99 Default 5 If not transmitted, the default value will be used to overwrite

- Note that the high breathing and heart rate threshold should be greater than the low breathing and heart rate threshold, otherwise the setting will fail.
- Unless there is a clear purpose, it is not recommended to change the above default parameters.

Return to content

Property	Type	Required	Illustrate
code	String	Y	" 200 " indicates success " 500 " means failure " 777 " means offline
msg	String	Y	
data	Map	Y	See the table below

5.4. Data card API (4G device specific)

5.4.1. getPhoneNumber (query 4G card number information)

Functional Description

Query the card number information of the 4G device through the device UID.

Interface restrictions

⚠Only supported on 4G devices .

Request method

HTTP GET application/x-www-form- urlencoded

Request address

https://qinglanst.com/prod-api/thirdparty/v2/getCallee

Request Parameters

Parameter name	Type	Required	Value Description
uid	String	yes	Device UID

Return to content

Property	Type	Required	Illustrate
code	String	Y	" 200 " indicates success " 500 " failed
msg	String	Y	
data	Map	Y	See the table below

Details of data

Property	Type	Required	Illustrate
mobile	String	Y	telephone number

Example

Return Parameter

```
{
     " code " : " 20 0" ,
      " msg " : " success " ,
    " data " : {
          "mobile": "14873666227"
        }
}
```

5.4.2. getCardInfo (query 4G card traffic information)

Functional Description

Query the remaining traffic and other information of the 4G device through the device UID.

Interface restrictions

⚠Only supported on 4G devices .

Request method

HTTP GET application/x-www-form- urlencoded

Request address

https://qinglanst.com/prod-api/thirdparty/v2/cardInfo

Request Parameters

Parameter name	Type	Required	Value Description
uid	String	yes	Device UID

Return to content

Property	Type	Required	Illustrate
code	String	Y	" 200 " indicates success " 500 " failed
msg	String	Y	
data	Map	Y	See the table below

Details of data

Property	Type	Required	Illustrate
residualFlow	String	Y	Remaining traffic, in MB
totalFlow	String	Y	Monthly package traffic, in MB
status	String	Y	Return card status text: -1 Error data 1: Can be activated 2: Activated 3: Deactivated 4: Illegal suspension 5: Expired 6: Can be tested 7: Operator cancellation 8: User cancellation 9: Under-suspension 10: Report loss 11: Inventory 12: Faulty card 13: Disconnected 14: Unregistered suspension 15: Pre-cancellation 16: Excessive suspension 17: Suspended to keep number 18: Locked 19: Partially deactivated - Aggregate card 20: Unregistered network
expiredTime	String	Y	Expiration time

Example

Return Parameter

{
"msg": "Operation successful",
"code": 200,
"data": {
" residualFlow ": "482.0", -- Residual flow (in MB)
" totalFlow ": "500", --Total flow this month (in MB)
"status": " Activated ", --Card status: -1 Error data 1: Can be activated 2: Activated 3: Deactivated 4: Illegal suspension 5: Expired 6: Can be tested 7: Operator cancellation 8: User cancellation 9: Under suspension 10 : Report loss 11: Inventory 12: Faulty card 13: Disconnected 14: Unregistered suspension 15: Pre-cancellation 16: Excessive suspension 17: Suspended to keep number 18: Locked 19: Partially deactivated - Aggregate card 20: Unregistered network
" expiredTime ": "2024-10-31 23:59:59" --Expiration time
}
}

5.4.3. getWhitelist (query whitelist)

Functional Description

Query the call whitelist of 4G devices by device UID. Only mobile phone numbers on the whitelist are allowed to call the device.

Interface restrictions

⚠Only supported on 4G devices .

Request method

HTTP GET application/x-www-form- urlencoded

Request address

https://qinglanst.com/prod-api/thirdparty/v2/whiteList

Request Parameters

Parameter name	Type	Required	Value Description
uid	String	yes	Device UID

Return to content

Property	Type	Required	Illustrate
code	String	Y	" 200 " indicates success " 500 " failed
msg	String	Y	
data	Map	Y	See the table below

Details of data

Property	Type	Required	Illustrate
whiteNumber	String	Y	Whitelisted mobile numbers
whiteNumberName	String	N	Name, no effect at the moment
status	String	N	The status currently returned is open

5.4.4. addWhitelist (Add a whitelist)

Functional Description

Add a whitelist for 4G devices by device UID. Only mobile numbers on the whitelist are allowed to call the device.

Interface restrictions

⚠Only supported on 4G devices .

⚠Each device supports a maximum of 5 whitelists .

⚠Due to the influence of the card vendor, there is a high probability that adding to the whitelist will fail. After failure, the card vendor will automatically enter the queue for a second addition. If you fail multiple times, please contact Hobacare

Request method

HTTP POST application/ json

Request address

https://qinglanst.com/prod-api/thirdparty/v2/whiteAdd

Request Parameters

Parameter name	Type	Required	Value Description
uid	String	yes	Device UID
mobile	String	yes	Phone number

Return to content

Property	Type	Required	Illustrate
code	String	Y	200 - Success 500 - Failure
msg	String	Y	
data	Map	Y	See the table below

Example

Return Parameter

```
{
    " code " : " 20 0" ,
    " msg " : " success " ,
}
```

5.4.5. delWhiteList (Delete Whitelist)

Functional Description

Delete the whitelist for a 4G device by device UID. After the whitelist number is deleted, the device will no longer be able to be called.

Interface restrictions

⚠Only supported on 4G devices .

⚠Only one whitelist can be deleted per card per month. Please do not add or delete irrelevant personnel frequently .

⚠Due to the influence of the card dealer, there is a high probability that the deletion of the whitelist will fail. After the failure, the card dealer will automatically enter the queue for a second addition. If it fails multiple times, please contact Hobacare

Request method

HTTP POST application/ json

Request address

https://qinglanst.com/prod-api/thirdparty/v2/whiteDelete

Request Parameters

Parameter name	Type	Required	Value Description
uid	String	yes	Device UID
mobile	String	yes	Phone number

Return to content

Property	Type	Required	Illustrate
code	String	Y	200 - Success 500 - Failure
msg	String	Y	
data	Map	Y	See the table below

Example

Return Parameter

```
{
    " code " : " 20 0" ,
    " msg " : " success " ,
}
```

5.5. Get the report

5.5.1. getSleepReport (Get Sleep Report)

Functional Description

Query the sleep report of the device on a certain date by device UID.
Note that the sleep report content is in Chinese by default. If you need to use English, please refer to 5.1.3 Setting HTTP Headers.

Interface restrictions

⚠The maximum call frequency is 20 times within 10 seconds. If it exceeds the limit, the current limit will be triggered. Hobacare only saves reports for 14 days by default.

Request method

HTTP POST application/ json

Request address

https://qinglanst.com/prod-api/thirdparty/v2/report

Request Parameters

Parameter name	Type	Required	Value Description
uids	Array<String>	yes	Device uid list, supports getting sleep reports of up to 5 devices at the same time
date	String	yes	Date yyyy -MM-dd
version	String	no	2.0 If not specified, default is 2.0 2.1 Added eye movement status

Return to content

Property	Type	Required	Illustrate
code	String	Y	" 200 " indicates success " 500 " means failure " 712 " indicates illegal operation
msg	String	Y	
data	Map	Y	See the table below

Details of data

Property	Type	Required	Illustrate		
uid	String	Y	Device serial number		
date	String	Y	date		
score	Number	Y	Sleep score - max 100		
fallScore	Number	Y	Fall score - maximum 35 points		
sleepEfficiency	Number	Y	Sleep efficiency		
sleepQuality	Number	Y	Sleep quality		
sleepTotalTime	String	Y	Total sleep time (min)		
onBedTime	String	Y	Total time in bed		
scoreLabel	String	Y	Sleep evaluation		
leaveBedCount	Number	Y	Number of times leaving bed at night		
getBedIdx	String	Y	Bedtime		
sleepStIdx	String	Y	Sleep start time		
sleepEdIdx	String	Y	Sleep end time		
leaveBedIdx	String	Y	Time out of bed		
ahi	String	Y	Respiratory disturbance index 0<times<=5 Mild 5<number<=15 Moderate >15 times Severe		
breathRateVo	Obj	Y	min	number	Minimum
			avg	number	average value
			max	number	Maximum
			avgLabel	string	evaluate
			dataList	array	Breathing data throughout the

					night
heartRateVo	Obj	Y	min	number	Minimum
			avg	number	average value
			max	number	Maximum
			avgLabel	string	evaluate
			dataList	array	All-night heart rate data
sleepIndexCommonList	Array	Y	name	string	Statistical name - meaningless
			value	string	Duration - minutes
			status	string	0-Deep sleep duration 1-Light sleep duration 2-Night awakening duration 3- Time out of bed at night 5-Time falling asleep 7-Eye movement duration
			ratio	number	neglect
			comment	string	neglect
statisticalData	Array	Y	startTime	string	Start time
			endTime	string	End time
			status	string	0-deep sleep 1-shallow sleep 2-awake 3-out of bed 4-offline 7-eye movement
timestamps	Array	Y	Timestamp array - string		
alarmEvents	Obj	Y	ts	string	time
			eventName	string	Event Name - No Meaning
			eventType	number	11-Hyperpnea 12-Hypopnea 13-Apnea 14-High heart rate 15-Low heart rate
evaluation	Obj	Y	sleepDeepRatioEvaluation	string	Deep sleep percentage -qualified
			sleepLeaveBedEvaluation	string	Number of times leaving bed at night - whether qualified
			sleepStartTimeEvaluation	string	Start sleeping time - Is it qualified?
			sleepDurationEvaluation	string	Sleep duration - whether it is adequate
			sleepAHIEvaluation	string	AHI indicator--Analysis
			ahiUnqualifiedEvaluation	string	ahi--qualified
			ahiAnalysisEvaluation	string	ahi exception

					description - empty if none
			unqualifiedEvaluation	string	Number of times leaving bed, duration of sleep, percentage of deep sleep
			sleepAnalysisEvaluation	string	Overall analysis
userActivity	Obj	N	entryRoomCount	number	Number of entries and exits
			inRoomDuration	string	Total time in the room (HH: mm: ss)
			otherDuration	string	Other durations (HH:mm:ss)
			otherDurationRatio	string	Other duration ratio
			speed	string	Speed m/min
			staticDuration	string	Static duration (HH: mm: ss)
			staticDurationRatio	string	Static duration ratio
			stepNumber	number	Steps
			walkDuration	string	Walking time (HH: mm: ss)
			walkDurationRatio	string	Percentage of walking time

Example: **Double click to download**



sleep report (English).txt

5.5.2. getRiskRanking (Get the ranking list)

Functional Description

Check the risk ranking of this account (large screen ranking).

Interface restrictions

⚠The maximum call frequency is 20 times within 5 seconds. If it exceeds the limit, the current limit will be triggered.

Request method

HTTP POST application/ json

Request address

https://qinglanst.com/prod-api/thirdparty/v2/riskRanking

Request Parameters

Parameter name	Type	Required	Value Description
			No parameters

Return to content

Property	Type	Required	Illustrate
code	String	Y	" 200 " indicates success " 500 " means failure
msg	String	Y	
data	Obj	Y	See the table below

Details of data

Property	Type	Required	Illustrate	Parameter			
fallAlarmRaking	Array	Y	Falling leaderboard	oldManName	String	no	name
				score	Number	yes	score
				eqtName	String	no	Device Name
				uid	String	yes	Serial Number
				age	Number	no	age
				enjoyDays	Number	no	Use days
				gender	String	no	0-Male 1-Female
aggregateScoreRaking	Array	Y	Comprehensive ranking	Same as above			

Example:


```
{
"msg": "Operation successful",
"code": 200,
"data": {
" fallAlarmRaking ": [
{
" oldManName ": "d2333",
"score": 100,
" eqtName ": "D233",
" uid ": "AD8A613BD233",
"age": 54,
" enjoyDays ": 212,
"gender": "0"
}
```

```
],
"  aggregateScoreRaking  ": [
{
" oldManName ": "d2333",
"score": 100,
" eqtName ": "D233",
" uid ": "AD8A613BD233",
"age": 54,
" enjoyDays ": 212,
"gender": "0"
},
{
" oldManName ": "test",
"score": 0,
" eqtName ": "radar",
" uid ": "CBAA7C29E9DA4DA",
"age": 0,
" enjoyDays ": 204,
"gender": "2"
},
{
" oldManName ":  null  ,
"score": 0,
" eqtName ": "8418",
" uid ": "CB262728F788418",
"age":  null  ,
" enjoyDays ":  null  ,
"gender": "0"
}
]
}
}
```


5.6. Get a token

5.6.1. getToken

Functional Description

If you want to use the embedded page, you must first obtain an access token.

Interface restrictions

⚠ Do not call repeatedly. The token you get each time is the same if it has not expired. Please save the cache .

Request method

HTTP POST application/ json

Request address

https://qinglanst.com/prod-api/login

Request Parameters

Property	Type	Required	Illustrate
username	String	yes	Refer to Section 2.1 to obtain
password	String	yes	
pattern	String	yes	monitor fixed value
grantType	String	yes	passwordFixed value

Return to content

Property	Type	Required	Illustrate
code	String	Y	Message code 200 Success Message code 500 failed
msg	String	Y	information
data	obj	Y	data

Data content:

Parameter name	Type	Required	Illustrate
access_token	String	Y	Token
refresh_token	String	Y	Used to refresh token, required for nested pages
token_type	String	Y	http request prefix
expires_in	number	Y	Expiration time in seconds

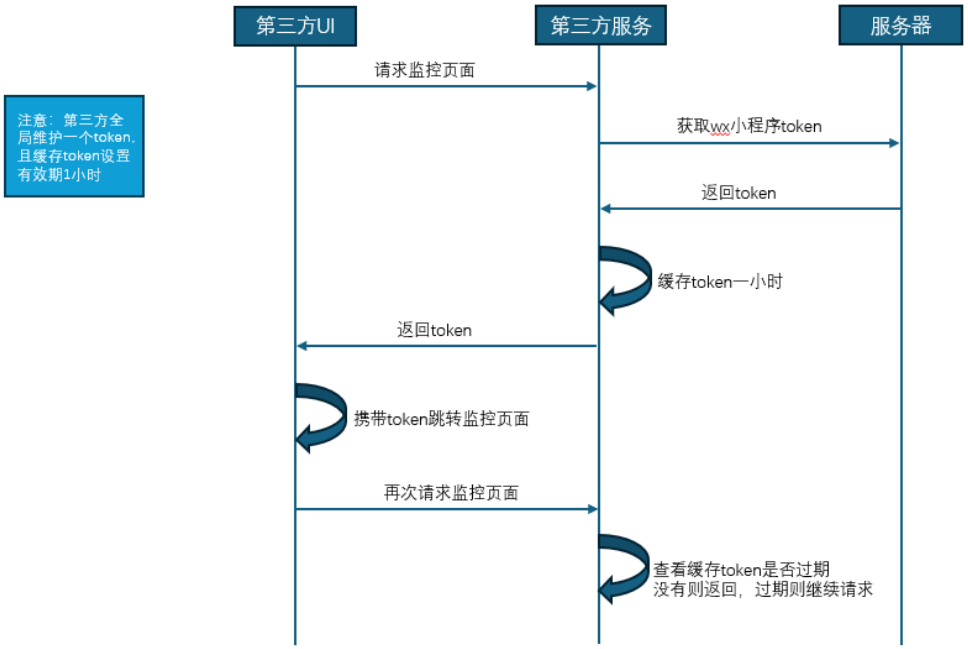
Example

```
Return to content
{
    " code " : " 20 0" ,
    " msg " : " success " ,
    " data " : {" access_token ": "82897602-b715-4b7c-a173-e9c507245a4a",, " token_type ": "bearer",, "
    expires_in ": 6576} //Expiration time in seconds }
```

5.6.2. getWxToken (Get mini program token)

Functional Description

This interface is used in conjunction with the H5/jump page in Section 4.2. Its usage is shown in the following flow:



Interface restrictions

⚠ Hobacare server maintains two tokens A and B for each enterprise account for cyclic use. When this interface is called for the first time, A token will be returned, B token will be used for the second call, and A token/B token will be used cyclically for the third/fourth call.

⚠The token is valid for 2 hours. Please request a refresh after one hour of use (if necessary). The previous token is still valid.

⚠Do not obtain tokens repeatedly in a short period of time. The interface is limited to once an hour.

Request method
HTTP POST application/ json

Request address
https://qinglanst.com/prod-api/ thirdparty/v2/applet/ a uth

Request Parameters
none
Return to content

Property	Type	Required	Illustrate
code	String	Y	Message code 200 Success Message code 500 failed
msg	String	Y	information
data	obj	Y	data

Data content:

Parameter name	Type	Required	Illustrate
access_token	String	Y	Token
refresh_token	String	Y	Used to refresh token, not supported yet
token_type	String	Y	http request prefix
expires_in	number	Y	Expiration time in seconds

Example

```
Return to content
 {
      " code " : " 20 0" ,
      " msg " : " success " ,
      " data " : {" access_token ": "82897602-b715-4b7c-a173-e9c507245a4a",, " token_type ": "bearer",, "
      expires_in ": 6576} //Expiration time in seconds }
```

5.7. MQTT Subscription

5.7.1. Enable MQTT push

Functional Description

The device pushes all the trajectory, breathing and heart rate data to a third-party server

Interface restrictions

⚠Since data will be pushed continuously, if you use a 4G device, please pay attention to the traffic limit

Request method

HTTP POST application/ json

Request address

https://qinglanst.com/prod-api/thirdmqtt/v2/dept/opmqtt

Request Parameters

Parameter name	Type	Required	Value Description		
uid	String	yes	Device UID		
messageType	Array Number	yes	Example [1,2,3,4] -- all open Analysis: 1-Second-level trajectory 2-second breathing heart rate 3- Minute-level trajectory statistics----counting people, steps, speed 4- Minute-level sleep statistics----average breathing and heart rate, deep sleep and light sleep		
topics	obj	yes	pub	String Array	MQTT push topic, currently only supports specifying one topic
			sub	String Array	Optional, not used
username	String	yes	MQTT Username		
password	String	yes	MQTT password		
clientId	String	yes	Client ID, it is recommended to use device UID, different devices should not be repeated		

keepAlive	Number	yes	MQTT heartbeat time, it is recommended to be no less than 60, otherwise it may be easily disconnected
port	Number	yes	MQTT server port
server	String	yes	MQTT server IP/domain name
protocol	Number	no	1-mqtt 2-mqtts does not transmit the default mqtt

Example:

{
" uid ": "9D8A32CEBA1B",
"server": "192.168.1.188",
"password": "sun123",
" clientId ": "2024-06-13",
" keepAlive ": 60,
" messageType ": [
"1",
"2"
],
"port": 1883,
"topics": {
"pub": [
"/topic/pub/sun/device"
]
},
"username": " sun_test "
}

Return to content

Property	Type	Required	Illustrate
code	String	Y	" 200 " indicates success " 500 " failed
msg	String	Y	

Example

Please refer to section 2.4.1

5.7.2. Disable MQTT push

Functional Description

Turn off MQTT push

Interface restrictions

⚠none

Request method

HTTP POST application/ json

Request address

https://qinglanst.com/prod-api/thirdmqtt/v2/dept/clmqtt

Request Parameters

Parameter name	Type	Required	Value Description
uid	String	yes	Device UID
messageType	String	yes	Example [1,2,3,4] -- all off Analysis: 1-second trajectory 2-second breathing heart rate 3- Minute-level trajectory statistics----used to count the number of people, steps, and speed 4- Minute breathing and heart rate -- used to generate sleep reports

Return to content

Property	Type	Required	Illustrate
code	String	Y	" 200 " indicates success " 500 " failed
msg	String	Y	

6. Appendix

Java signature example: double click to download



SignUtil.java